



UNIVERSITY OF LIEGE

2023-2024

**High-dimensional statistics
2023-2024**

**Project 1: Exploratory data analysis and
correlation analysis**

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Statistical analysis

1 Description of the context and of the potential interest of the corresponding data analysis

In order to carry out an exploratory analysis of the results of national assessments for secondary and university education of engineering students, a dataset is extracted from the databases of the Colombian Institute for Educational Evaluation (ICFES), which will form the basis of our results on this project. It contains information on 252 Colombian students and 21 variables, some of which are academic and others social. The structure of the data allows us to observe the influence of social variables and the evolution of students' learning skills. The quantitative variables, of which there are twelve, are mainly academic variables, based on the academic results of Colombian students during their lives as secondary school and university students. Meanwhile qualitative variables are mainly represented by social variables. They are listed in the table below and notice that each variable with the extension "_S11" is related to secondary education outcomes, and those with the extension "_PRO" are related to university education.

Quantitative variables : CC_S11: Citizen Competencies, BIO_S11: Biology, PEOPLE_HOUSE: number of people living in the same household as the student, MAT_S11: Mathematics, ENG_S11: English, QR_PRO: Quantitative Reasoning, CR_PRO: Critical Reading, ENG_PRO: English, FEP_PRO: Formulation of Engineering Projects, WC_PRO: Written Communication.

Qualitative variables : GENDER : student's gender (F, M), JOB: binary variable indicating whether the student has a job on a regular basis while studying, INTERNET: binary variable indicating whether an Internet connection is available at home, TV: binary variable indicating whether a television is available at home, COMPUTER : binary variable indicating whether a computer is available at home, CAR: binary variable indicating whether a computer is available at home, SCHOOL_TYPE: type of high school, SCHOOL_NAT: nature of high school (PRIVATE, PUBLIC), ACADEMIC_PROGRAM :academic program to which a student belongs.

2 Exploratory data analysis

2.1 Statistical and graphical summary of the variables

The data set contains missing values. They are all linked to the social variables. This probably means that the researchers did not have access to information that is sometimes not compulsory to be provided by a student, such as the JOB variable. Thereby when dealing the missing value, one can apply simply case-deletion strategy which deletes rows with a missing value in a cell. The figures below show the two states of the dataset before and after applying the case deletion strategy, and we have now 248 observations in the data set (figure 1).

The first graphical summary of interest is the box plots. They identify outliers individually for each component and can also assess the distributions of data across various groups.

Through the box plots (figure 2), one observes much variability in the results of students among the quantitative variables of the data, particularly those concerning the assessment for secondary and university education. Indeed, the quantitative variables related to their assessment in high secondary school spread out almost in the same way. The largest F-spread of among them is one of the BIOS_11 variable and its corresponding value is : $F_u - F_l = 67.25$

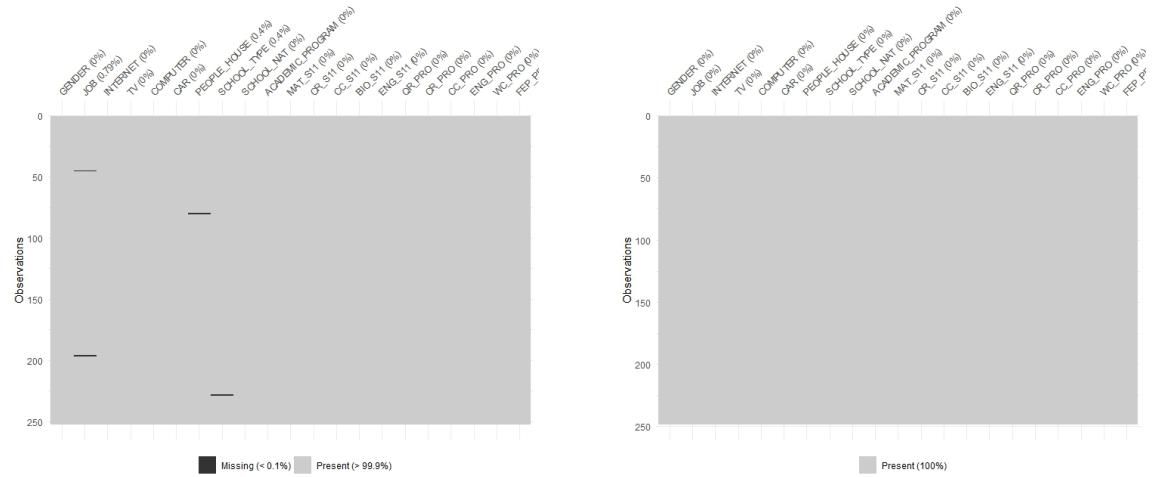


Figure 1: Wrangling missing values (left figure holding missing values and right figure without missing values)

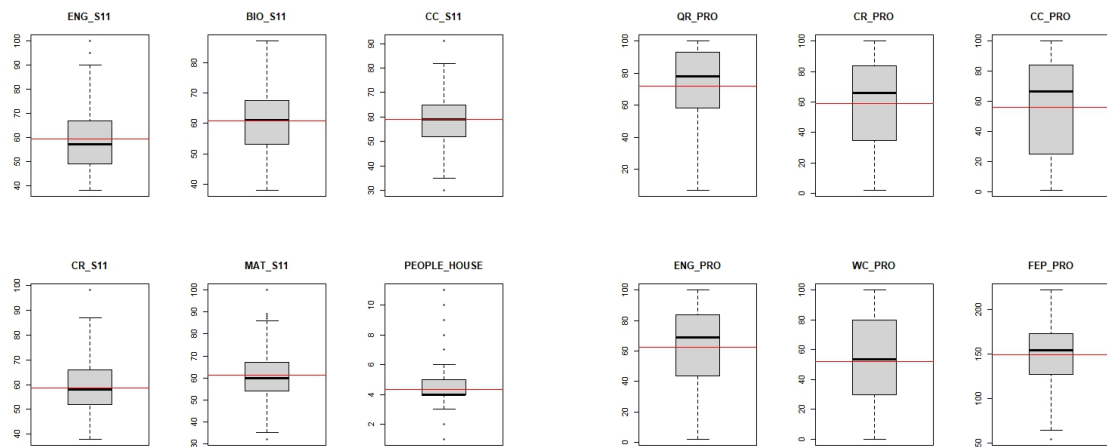


Figure 2: Box plots of variables highlighting the Five Number Summary (on the left figure those related to high school, and on the right, those related to university education).

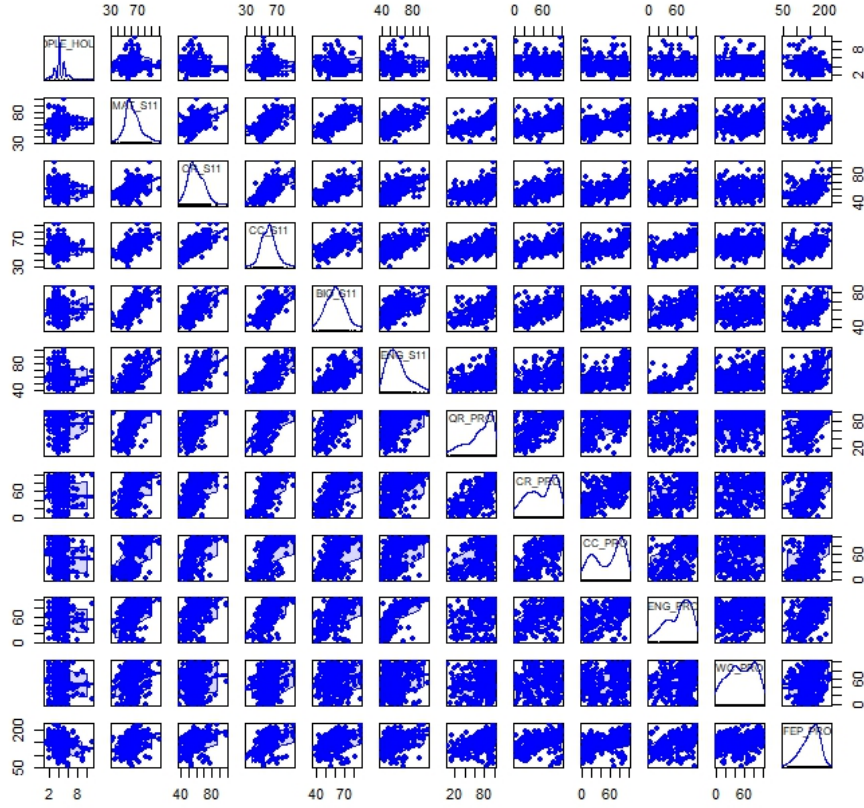


Figure 3: Scatter matrix of all quantitative variables

- 53 = 14,25. Meanwhile the quantitative variables related to their assessment in university education, provide us larger variability whose the largest F-spread is the one of the CC_PRO variable. Its corresponding value is 59 by applying the same previous formula. the are presence of outliers (non robust detection) in the BIO_S11, CR_S11 and MAT_S11 variables. This means there are students who obtained better results in those academic competencies than others, precisely two students obtained results which are too larger than the rest of students in Biology (BIO_S11) (the best result is 100), one student obtained the largest result than others in Critical Reading (CR_11) and we have the same observation for Mathematics (MAT_S11). These outliers are marked as a small circles. At the second moment of their student life, one can retain the following observation : all the variables related to their assessment in university education are skew since the lower half of the data (above the median) is more spread out than the upper half (below the median), the mean (as a non-robust measure of location) is pulled away from the median in CC_PRO, ENG_PRO, QR_PRO and CR_PRO variables. Red horizontal lines display the mean of each variable in a given box plot and it is used for illustrating how the mean is pulled away from the median in some variables.

Via the dispersion matrix, we have the density functions of each quantitative variable along its diagonal, and we can see that they are all unknown.

First of all in the figure 3, we find that the number of people at home has no influence on the

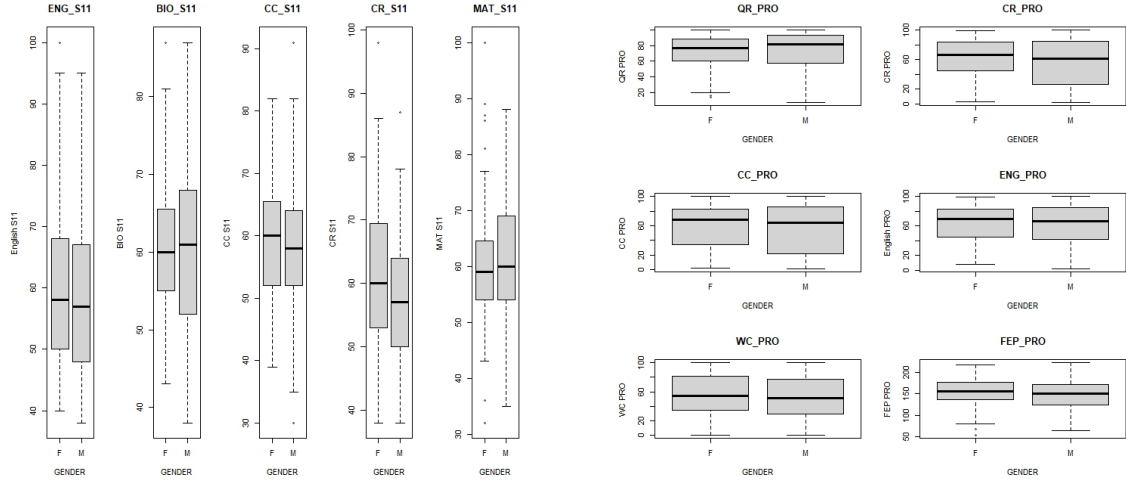


Figure 4: Box plots of variables highlighting the Five Number Summary of different groups of GENDER variable in the quantitative variables

student's assessment, since it is weakly correlated with one or other of the variables. There is little correlation between university teaching units. This is because they do not, for the most part, draw on the knowledge of others in their field, and more specific objectives in their choice of course.. This is reflected in graphs where the point clouds have no preferred direction and cover almost all the space reserved for their representation.

However, we note that the scatter plots for secondary education subjects are correlated. This is because these subjects require knowledge of others. For instance, CR_S11 and CC_S11 variables the two measures assess skills that are crucial for understanding and interacting with the social world and textual information. Hence the significant correlation between the two. So a Colombian student who does well in one of the two probably does well in the other. The same reasoning can be applied to other subjects in secondary education.

In addition, there is a link between certain subjects at secondary school and those at university, such as CC_S11 and CC_PRO Citizen Competencies, CR_S11 and CR_PRO for Critical Reading, and ENG_S11 and ENG_PRO for English. For CC_S11 and CC_PRO assess citizenship-related skills, CR_S11 and CR_PRO assess critical reading skills, and ENG_S11 and ENG_PRO evaluate the ability to communicate effectively in English. Hence the correlation between these pairs of variables, as shown in Figure 3.

2.2 Analysis of the potential association between some qualitative variables and the score-variables

At their first moment student life, the male gender students have gotten better outcomes for some subjects like English and Citizen Competencies than female gender students. On the other hand, female gender students were the best in mathematics and critical reading subjects. Moreover one sees some outliers (non robust detection) displaying the best results obtained by female gender on those both subjects. At their second moment student life, their median values in all subjects seem to be equal. All these remarks come from the figure 4.

The students who followed a university programme oriented towards industrial engineering were the ones who performed best in both assessments. We can identify this group of students

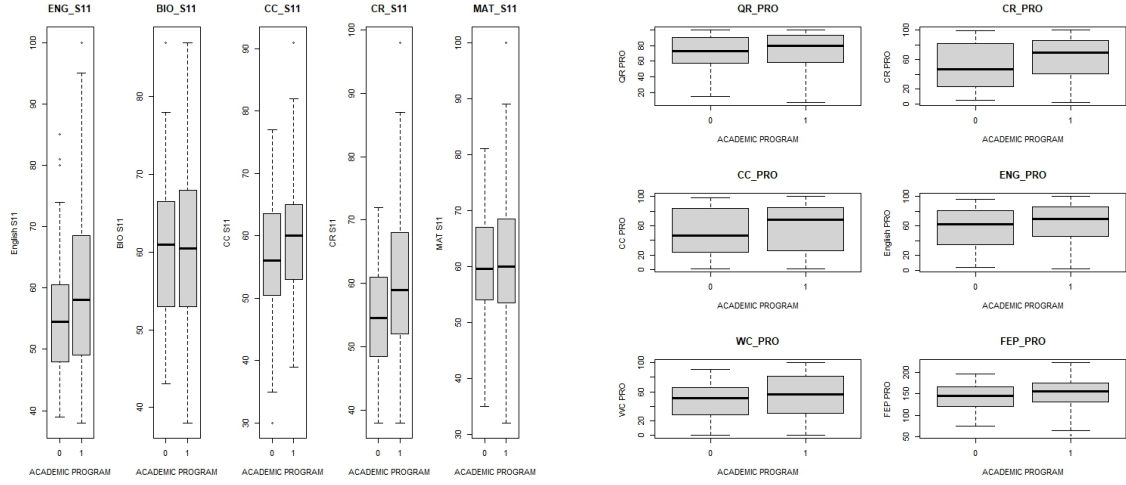


Figure 5: Box plots of variables highlighting the Five Number Summary of different groups of ACADEMIC_PROGRAM variable in the quantitative variables

in the box plots where it is marked 1, and 0 for those who have followed a mechatronics engineering programme. Comparing the median on the box plot for each subject for each group illustrates this previous reference in several box plot's subjects (figure 5).

2.3 Outlier detection based on the score-variables by comparing robust distances³ and classic Mahalanobis distances

Since the FEP_PRO score is out of 300 compared with the other subjects, we start by standardising our score-variables. Using the Mahalanobis distance (a non-robust detection technique), we see 27 observations above the threshold whose value is 18.307, represented by the red horizontal straight line in the graph below. This means the empirical correlation matrix is corrupted.

With the robust detection technique (FASTMCD), for quantile.used equal to 0.75 i.e. considering only the optimal subset of 0.75 corresponding to 186 observations on which we will deduce estimates such as that of the covariance matrix, we now obtain 16 observations which are outliers that stretch the dispersion of the points in the scatterplot. We therefore need to remove all these outliers from the data set in order to obtain good interpretations of our analysis (figure 6). The observations in the upper right frame, represent the outliers which are a number of 16, and those in the the upper left frame are the observations which are just masked. The lower left frame holds cleaned data (regular points).

The entries of the correlation matrix estimate derived from the application of the FASTMCD technique increase slightly compared to the entries of the non corrupted empirical correlation matrix. We also find that there are weak correlations between the WC_PRO variable and the other variables. Although this course requires knowledge of English, its main objective is to convey information, ideas or messages to an audience through the written word. Hence the uninteresting correlation with other English subjects. As mentioned above beyond via the scatter matrix (figure 7).

We see three sub-groups : the first with value pairs belonging to $[0;0.3[\times [0;0.3[$, the

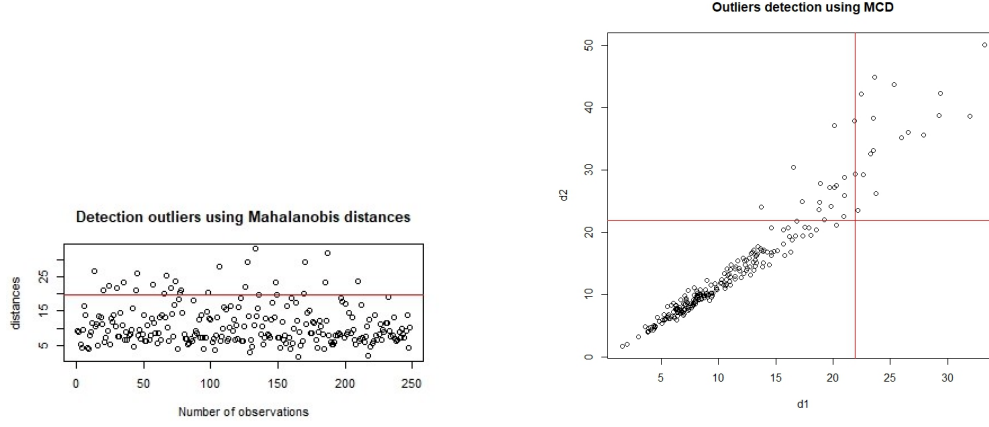


Figure 6: Outliers detection via Mahalanobis distances and FASTMCD respectively in the left and right figures (left figure holding missing values and right figure without missing values)

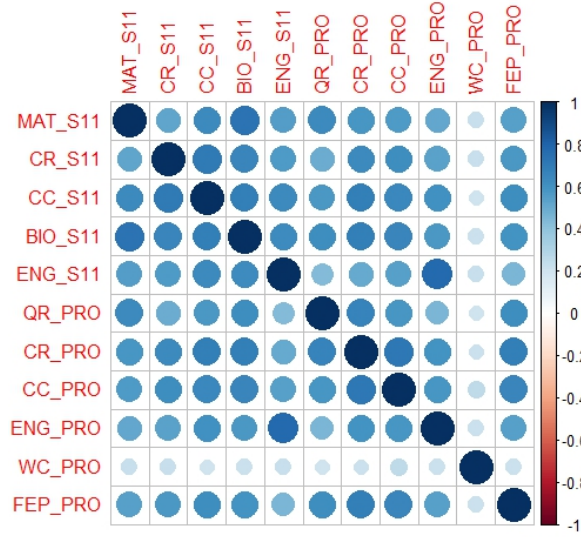


Figure 7: Pearson correlation matrix estimate from FASTMCD technique

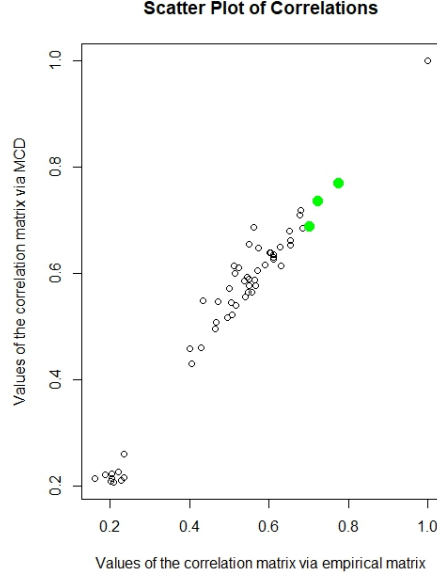


Figure 8: Scatter plot of pairs of values from the empirical and estimated correlation matrices

second to $[0.4;0.85]$ and the third represented by a single point with coordinates (1,1). The first cluster represents the correlation matrix entries related to WC_PRO variable with the data variables, and highlight the weak correlations with them. The second cluster illustrate the variables which are most correlated (figure 8).

3 Further exploration via a dimension reduction

On this part, the variable WC_PRO is dropped from the data set due to its weak correlations with others variables for the next. Via the cumulative percentage of explained variance attains at least 72.58 % with two first principal components. In the scree plot below, we can see that those both principal components that achieve more than 70% of a cumulative explained variance, are beyond the red horizontal line. It depicts each PC's associated to its proportion variance value multiplied by 10 (figure 9).

The two graphs (correlation circle and Pearson correlation matrix) in Figure 9 reveal significant correlations between the first principal component and the score variables. It can be seen that all the variables in the data are negatively correlated with the axis of the first principal component. The variables that contribute most negatively to the first component are BIO_S11, CR_S11 and CC_S11. Their correlation values with the second principal component are close to zero. One of them contains only the variables ENG_S11 and ENG_PRO, which have negative correlation values along the first and second principal components, and illustrates the strong correlation between the two, thanks to the use of English language skills acquired in secondary education, which is an extension of university. There are also two groups of clusters, the first made up of the ENG_S11 and ENG_PRO English disciplines, and the second made up of the others. We can interpret our correlation circle by adding that, success in studies is therefore mainly focused on subjects related to those of university education such as Citizen Competencies, Critical Reading and Biology; and knowledge of English subject is

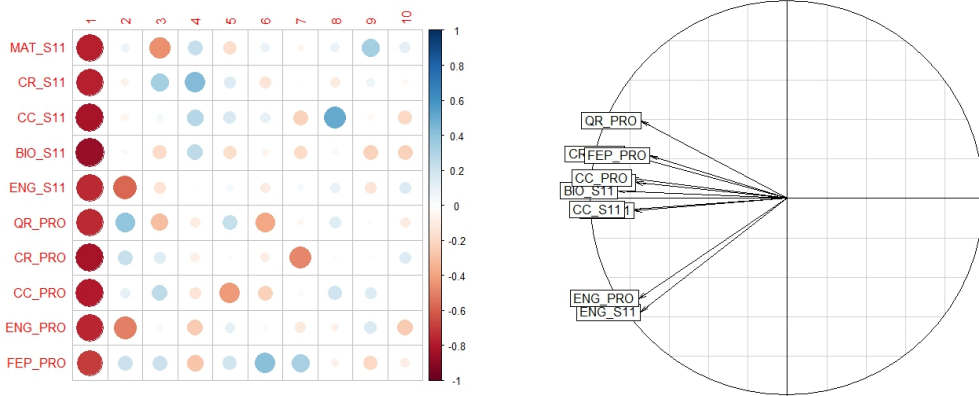


Figure 9: Pearson Correlation matrix between score variables and principal components on the left and the correlation circle on the right

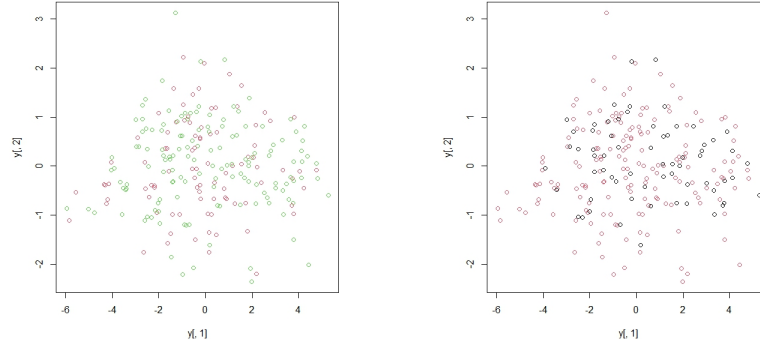


Figure 10: Construction of the projection but the groups defined by GENDER (in the left) and ACADEMIC_PROGRAM (in the right)

a plus.

Among the categorical variables, we will not investigate the TV, CAR, INTERNET or COMPUTER variables, for example. For reasons already mentioned, and moreover, these are not variables that contribute to student performance. For example, a student may not have a computer but the school provides a computer room, or a car because public transport exists. We will focus on two qualitative variables, namely GENDER and ACADEMIC_PROGRAM, for reasons already mentioned above.

We have an overlaying groups which makes it difficult to separate the groups in a reduced dimension space. This is due to non-linearity between some variables in the data (low correlations), like the subjects of university education which are not correlated between them. In the graphs below, the group of industrial engineering students is identified by red dots and the group of mechatronics students by black dots; and the group of male students is identified by green dots and the group of female students by black dots (figure 10).